RIT

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GA Progress Update – Filtered iCPursuitNeuralUCB Analysis & Friday Meeting

Daniel Krutz <dxkvse@rit.edu> Thu, Oct 16, 2025 at 5:44 AM
To: "Piter Garcia (RIT Student)" <pzg8794@rit.edu>

I haven't read through everything yet, but focus on your dad and sleep. This work is very very much secondary to your family. Family ALWAYS comes first.

Let's chat once things calm down with your father and you get rest. There's no need to chat tomorrow. Let's catch up next week if the situation allows.

I hope things go as well as possible.

Take care
-Dan

> On Oct 16, 2025, at 1:51 AM, Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:
>
> Hi Dan,
>
> I hope everything is well with you.
>
> I'm writing to provide a progress update as per our last meeting, and to discuss our Friday meeting.
>
> **Multi-Run Algorithm Performance Analysis (Filtered for Successful Runs):**
> I've completed comprehensive testing across 18 experimental notebooks, analyzing runs for our key models while excluding configurations with 0 rewards/efficiencies (resource failures to address later); I have a fix for this time of CPU resource allocation issue in the new environment, where I will be validating this further. This yields clear, reproducible patterns from viable outcomes, with **24 valid iCPursuitNeuralUCB runs**.
>
> **Overall iCPursuitNeuralUCB Multi-Run Performance (24 Valid Runs):**
>
> - Peak Oracle efficiency: **99.9%** (various frames)
> - Consistent ≥75% efficiency: 13/24 runs (**54.2%**)
> - Reliable ≥70% efficiency: 22/24 runs (**91.7%**)
> - Average efficiency: **77.6%**
> - Stable performance range: 55.9%-99.9% (70%+ in 91.7% of cases)

Final Run Patterns (Last/Highest Frame Efficiencies, Per-Model Filtered for >0%):
Note: Patterns derived from successful final runs per model (no per-model zeros), as no experiment had universally successful outcomes across all models without isolated failures.

- **iCPursuitNeuralUCB**:
  Average 81.4%, Range 75.3%-99.1%, Most Common: 70-80% (75% of successful final runs). Achieves
  100% ≥75%
  in filtered final runs, with 25% ≥90%. Validates previous findings (77.2% at 4000 frames) through ARIMA-enhanced predictive exploitation under proper resource management.

- **CPursuitNeuralUCB (Baseline)**:
  Average 60.9%, Range 17.3%-99.5%, Most Common: 60-80% (50% of successful final runs). Reduced variance post-filter; 33.3% ≥75%, 16.7% ≥90%, showing context adaptation limitations in sustained scenarios.

- **EXPNeuralUCB**:
  Average 64.7%, Range 41.9%-87.7%, Most Common: 60-80% (57% of successful final runs). Improved stability; 28.6% ≥75%, peaking at 87.7% when exploration converges, but prioritizing robustness over peak final exploitation.

Key Pattern: In successful configurations, iCPursuitNeuralUCB's hybrid contextual pursuit + informative prediction architecture delivers consistently superior final-run performance (**81.4% avg** vs. **64.7% EXPNeuralUCB**, **60.9% CPursuitNeuralUCB**), proving its advantage for reliable quantum entanglement routing. The filtered multi-run average of **77.6%** aligns with prior reports, emphasizing the model's strength when isolating resource issues.

Framework Development:
I've also developed multiple optimizations for quantum entanglement routing, including enhanced qubit allocation, realistic quantum environment modeling, and resource failure retry mechanisms. These await testing post-rest.

Personal Situation:
Due to a family emergency, I had to take my father to the ER, and he is still undergoing medical tests. I've been at the hospital for extended hours and have not slept in over 48 hours, preventing proper validation and documentation of findings. I might have to be at the hospital on Friday, but I can work remotely and join our meeting virtually or by phone if needed. I'll confirm by Thursday evening.

Next Steps:
After resting, I'll:

1. Validate and document the filtered multi-run iCPursuitNeuralUCB patterns

2. Address 0-reward failures with retry mechanisms
3. Test quantum environment optimizations
4. Integrate the feedback and template provided in your previous email.

I apologize for the delay and appreciate your understanding. The patterns from successful runs are robust and align with our prior reports.

Please let me know if you have questions.

Best regards,
Piter